



МОСКОВСКИЙ ГОСУДАРСТВЕННЫЙ
ТЕХНИЧЕСКИЙ УНИВЕРСИТЕТ
ИМ. Н.Э. БАУМАНА

A WHEELED AUTONOMOUS ROBOT WITH ALL-TERRAIN CAPABILITY FOR PATROLLING AND INSPECTING FORESTED AREAS.

Student

Artem Sannikov

CM7-21M

Scientific supervisor: Gulyaev I. A.





The forest area of the Moscow region is

1,883,329 hectares.

Length of forest protection routes:

400-800 km.

Hundreds of man-hours are spent
on forest inventory surveys.

RELEVANCE

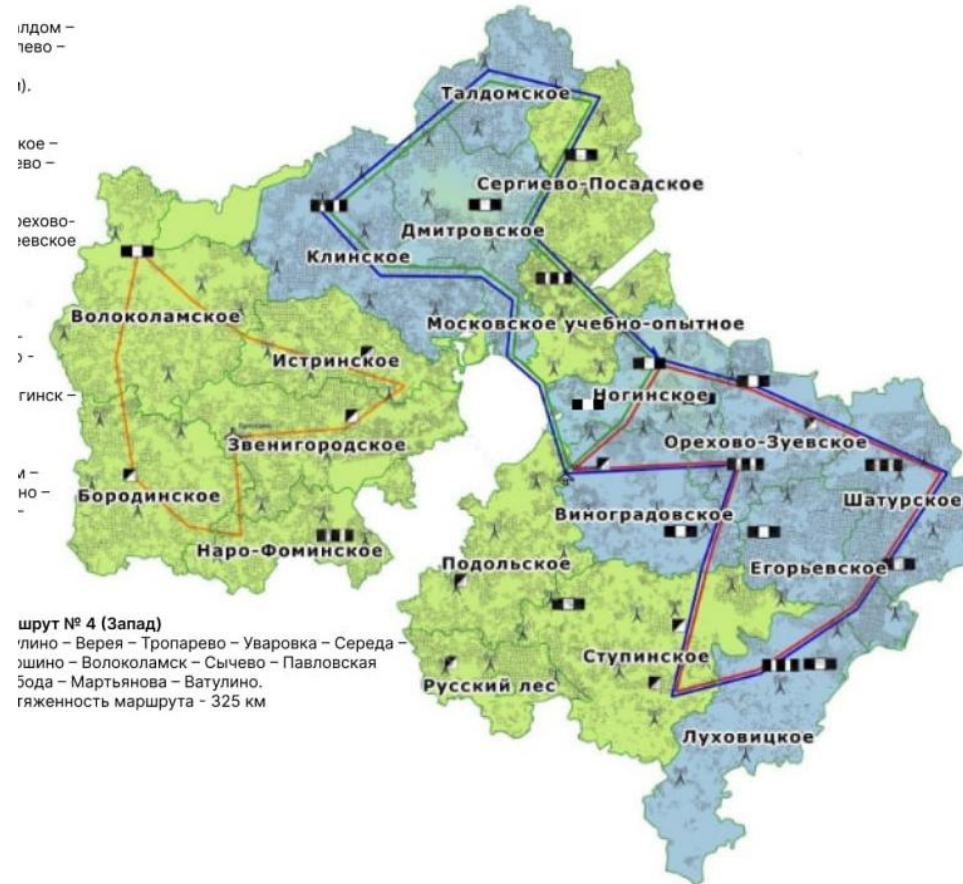


Fig. 1. Forest protection map by State Autonomous Institution of Forestry “Mosobles” Ministry





GOAL AND OBJECTIVES



Goal

Wheeled-walking platform kinematics and control system development

Objectives

- UAV feasibility assessment
- Kinematic design
- Motor selection
- Control system design
- Functional diagram development
- Mobility and control system evaluation





FUNCTIONAL DIAGRAM OF THE SYSTEM



- Designed for unpaved roads and rough terrain navigation
- Full area coverage not guaranteed in dense forests
- UAV proposed as a backup monitoring system

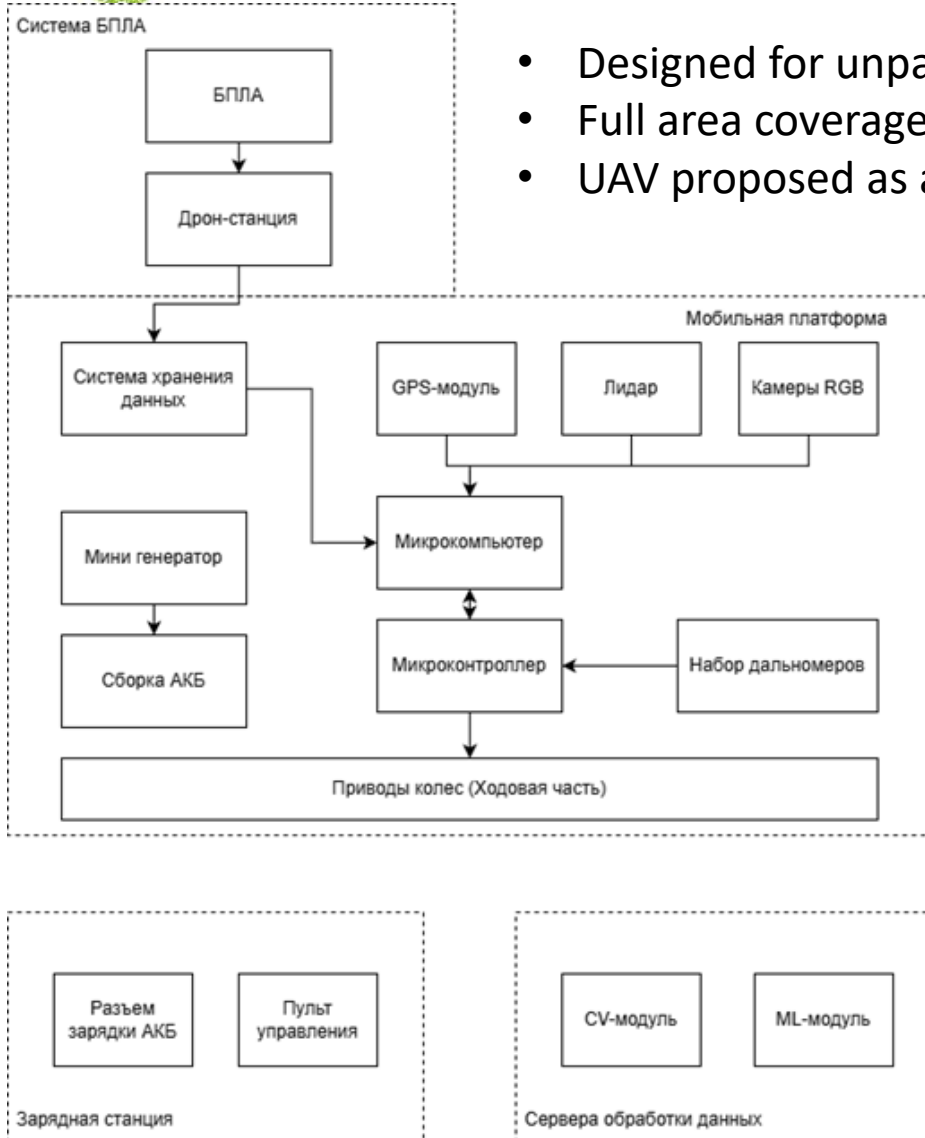


Fig. 2 Functional diagram

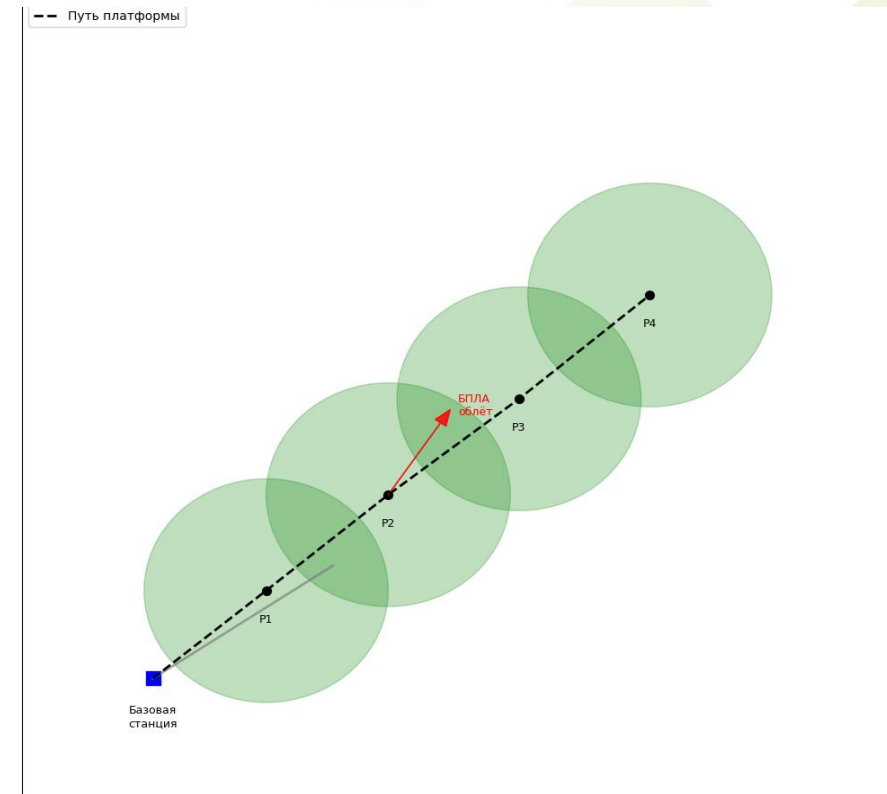


Fig. 3. Movement scheme and scanning of the area using UAVs



GENERAL STRUCTURE OF THE SYSTEM



The customer provided a specification with a list of the required functionality and requirements for the system, which was divided into several parts.

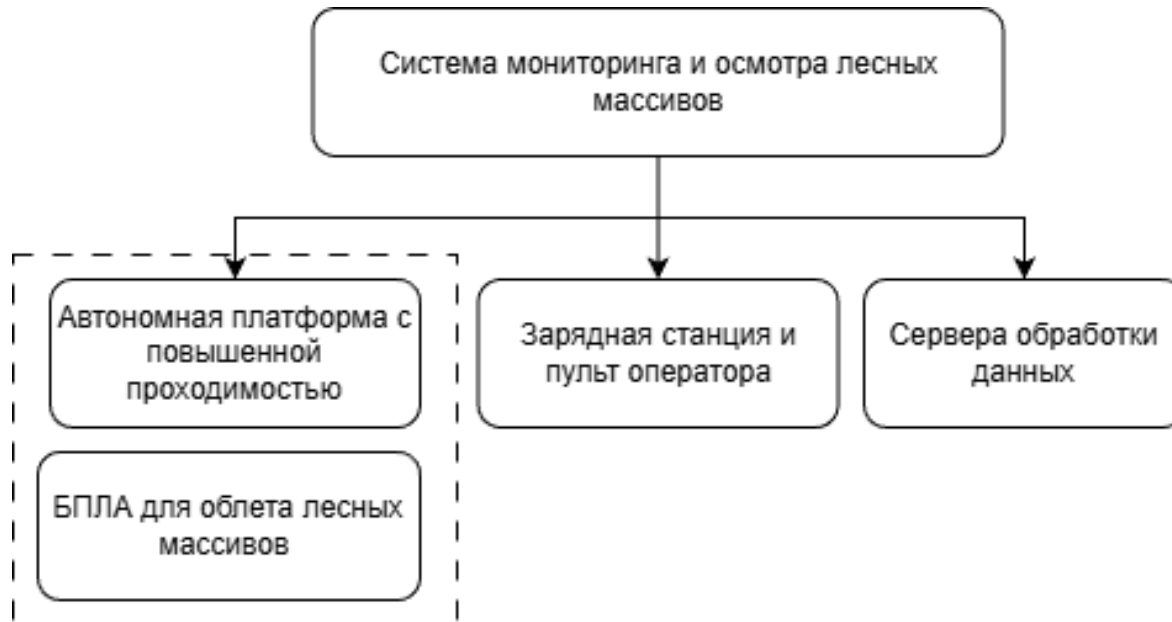


Fig. 4 System structure

The system's tasks include:

- Autonomous movement and territory survey
- Navigation, localization, and mapping
- Tree inventory and identification
- Data collection and reporting

My Goal:

Development of an autonomous wheeled platform



*UAV and drone station are used as a finished product



TRANSPORT PLATFORM MODEL

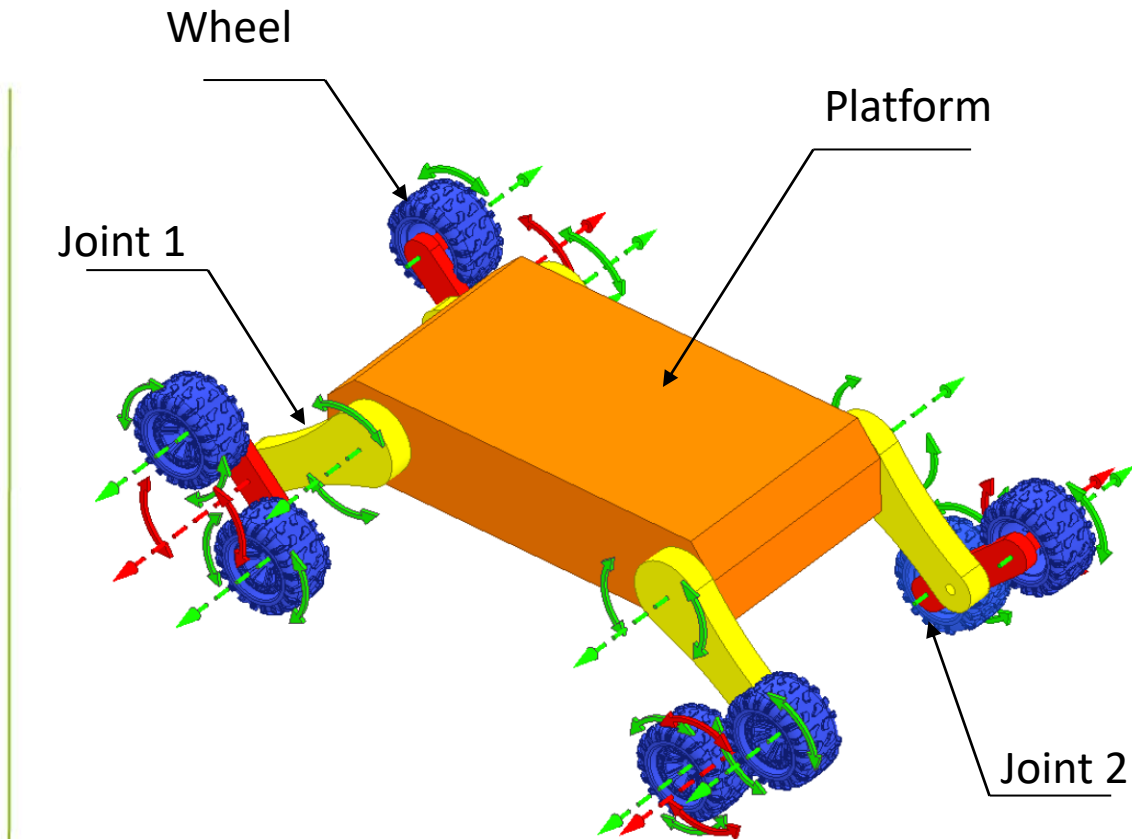


Fig. 5 Kinematics model, 16 axes
of rotation

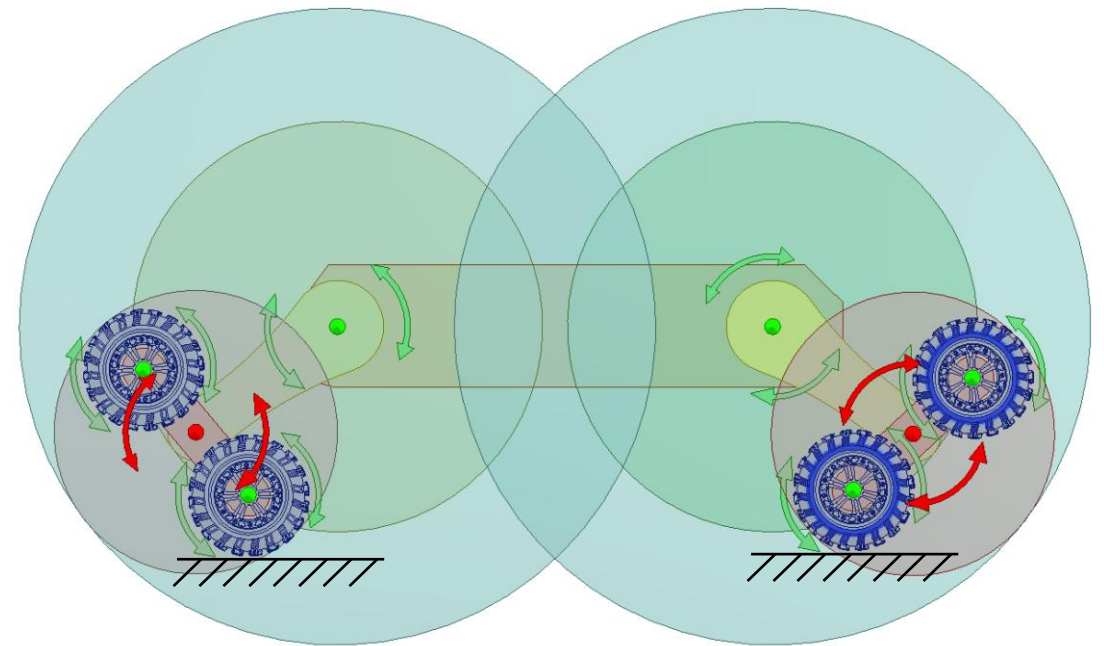


Fig. 6 Working area of each link





DIFFERENT STATES OF MOTION

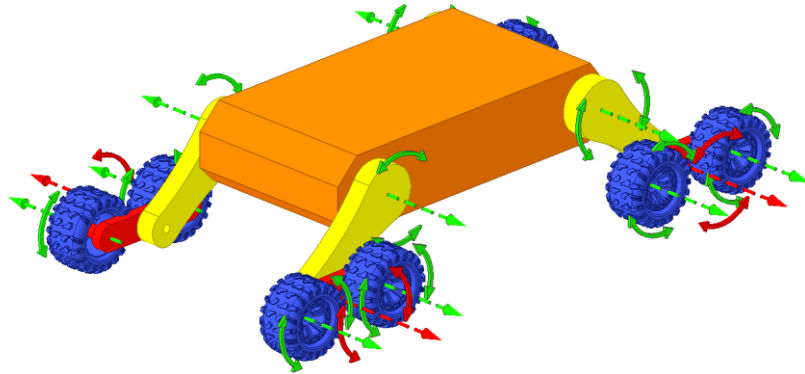


Fig. 9 Accelerated movement on a flat area

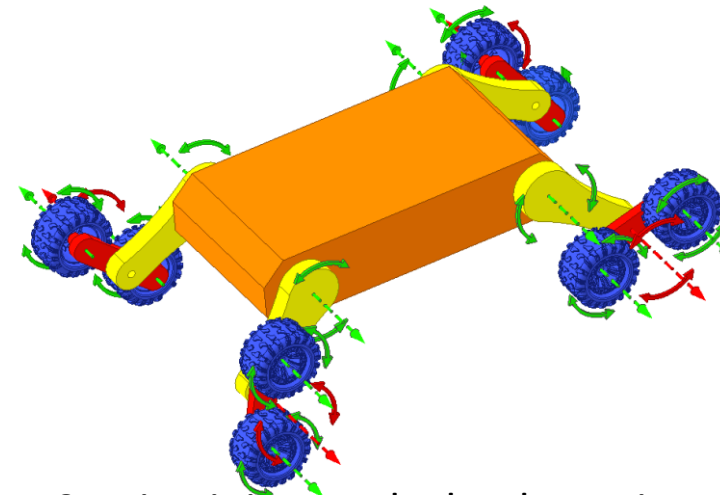


Fig. 10 Using joints and wheel rotation to overcome obstacles

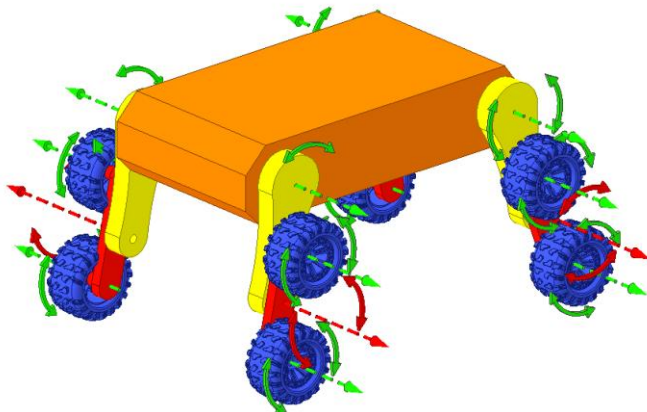


Fig. 7 Maximum ground clearance

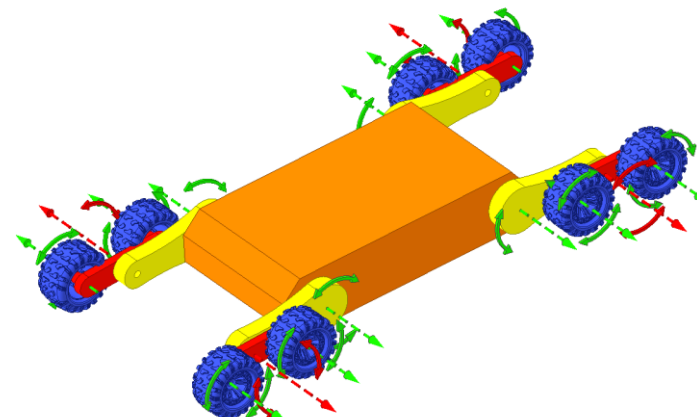


Fig. 8. Stable position
"On the belly"





EVALUATION OF THE OBTAINED PASSABILITY



- Ground pressure of **0.29 kg/cm²** with a robot weight of 170 kg and 8 wheels in contact simultaneously – reducing the pressure in the chambers reduces ground pressure.
- Max cross-country ability (theoretical): **339 mm**
- Min slope capability (including stabilization): **22°**

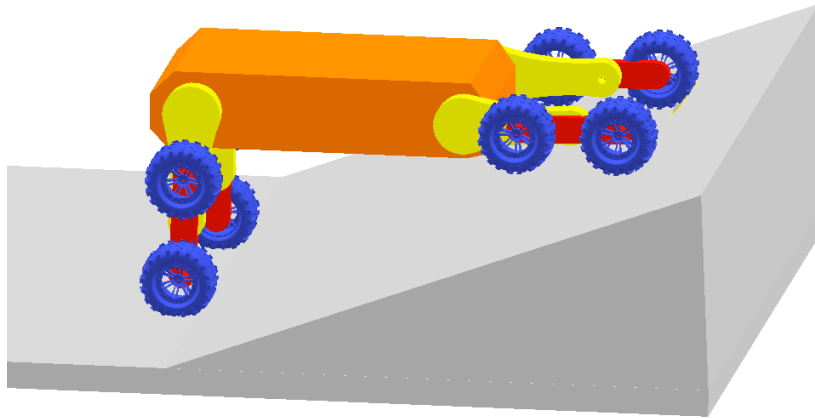


Fig. 12 Slope assessment

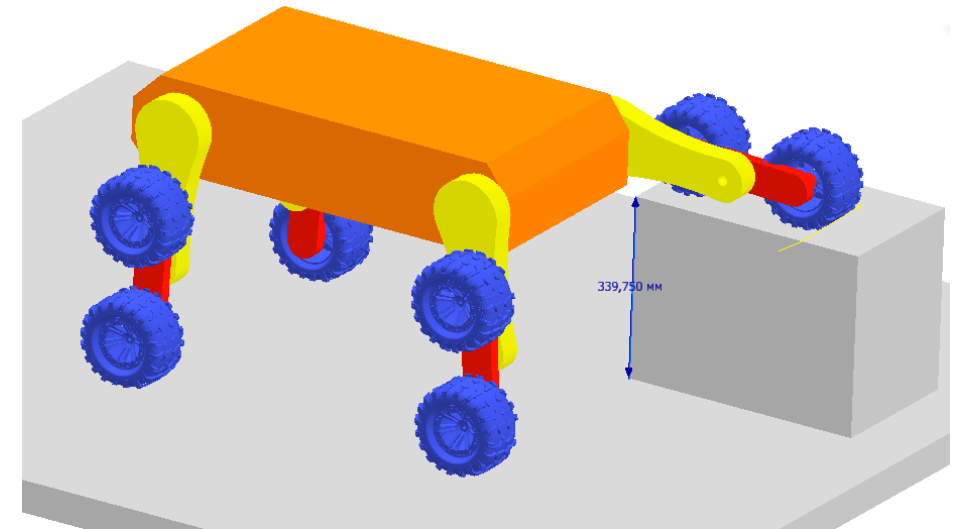


Fig. 11. Assessment of profile cross-country ability





ENERGY CALCULATION: FINDING DRIVE CAPACITIES



Drive power calculation for hip, knee and wheel joints

Power requirement for the Wheel :

$$(P_{дв})_p = \xi \frac{P_H}{\eta_0} = 1.2 \frac{204.69}{0.6} = \mathbf{408 \text{ Вт}}$$

Power requirement for the Knee:

$$(P_{дв})_p = \mathbf{1326 \text{ Вт}}$$

Power requirement for hip:

$$(P_{дв})_p = \mathbf{1862 \text{ Вт}}$$

drive	torque	Speed
Wheel	3.68 Н * м	55,6 рад\с
Knee	97 Н * м	6,8 рад\с
Hip	194 Н * м	5,8 рад\с

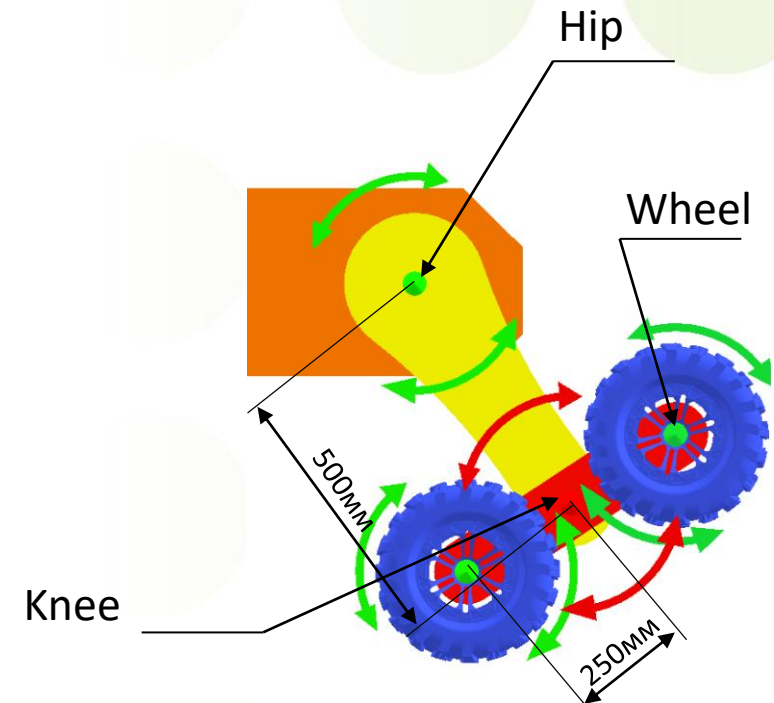


Fig. 13 Axes Designation



SELECTION OF ENGINES



KY215ACS0450-15
Servomotor for Hip
Drive



KY110AS0420-25 servo
motor for knee drive



KY165ACS0405-02B
Wheel Servo Motor for
Wheel Drive



KY40AS0105-12
Axis Locking Servo
Motor

Power	4000 Вт	1500 Вт	500 Вт	70 Вт
Speed	1500 об/мин	1500 об/мин	250 об/мин	63 об\мин ()
troque	31.8 Н*м	9.5 Н*м	4.2 Н*м	4 Н*м
voltage	48 В	48 В	48 В	12 В
Mass	12 кг	4 кг	3 кг	0.6 кг
encoder	Yes	Yes.	Yes	Yes 10



SIMSCAPE ROBOT CONTROL SYSTEM

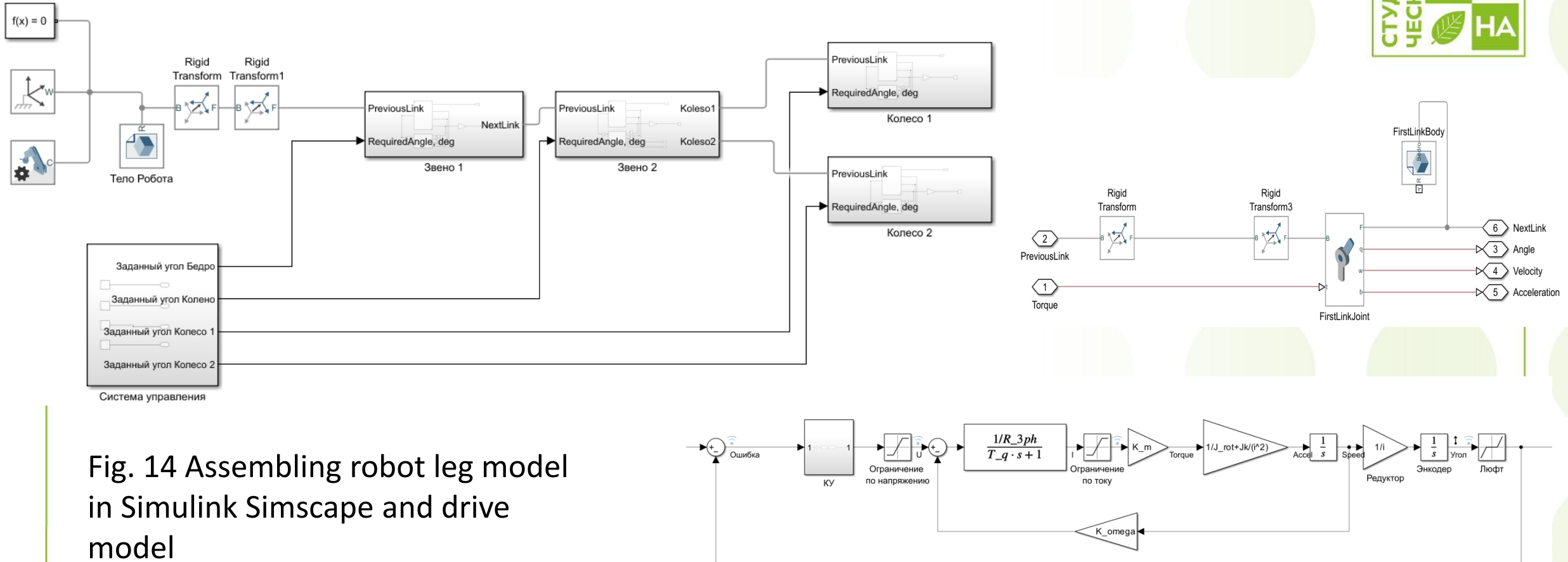


Fig. 14 Assembling robot leg model in Simulink Simscape and drive model

KY110AS0420-25 drive, gearbox $I = 23$ was used.



SIMSCAPE MULTIBODY MODELING

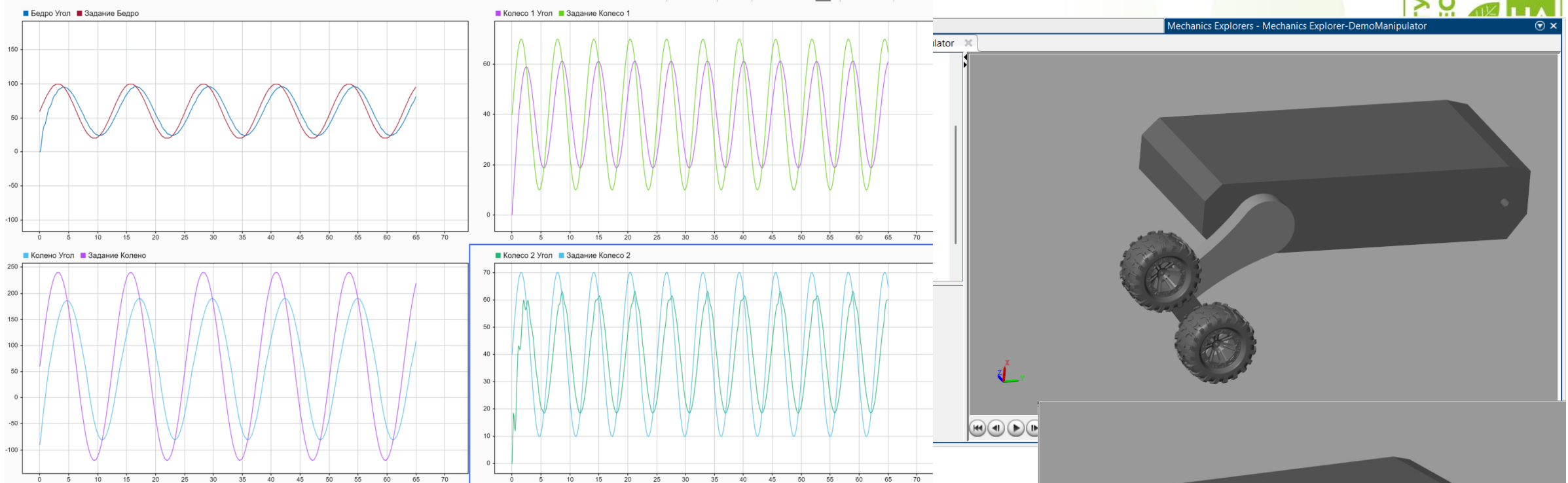


Fig. 15. Robot Animation in
Mechanics Explorer





FUNCTIONAL DIAGRAM OF THE TRANSPORT PLATFORM

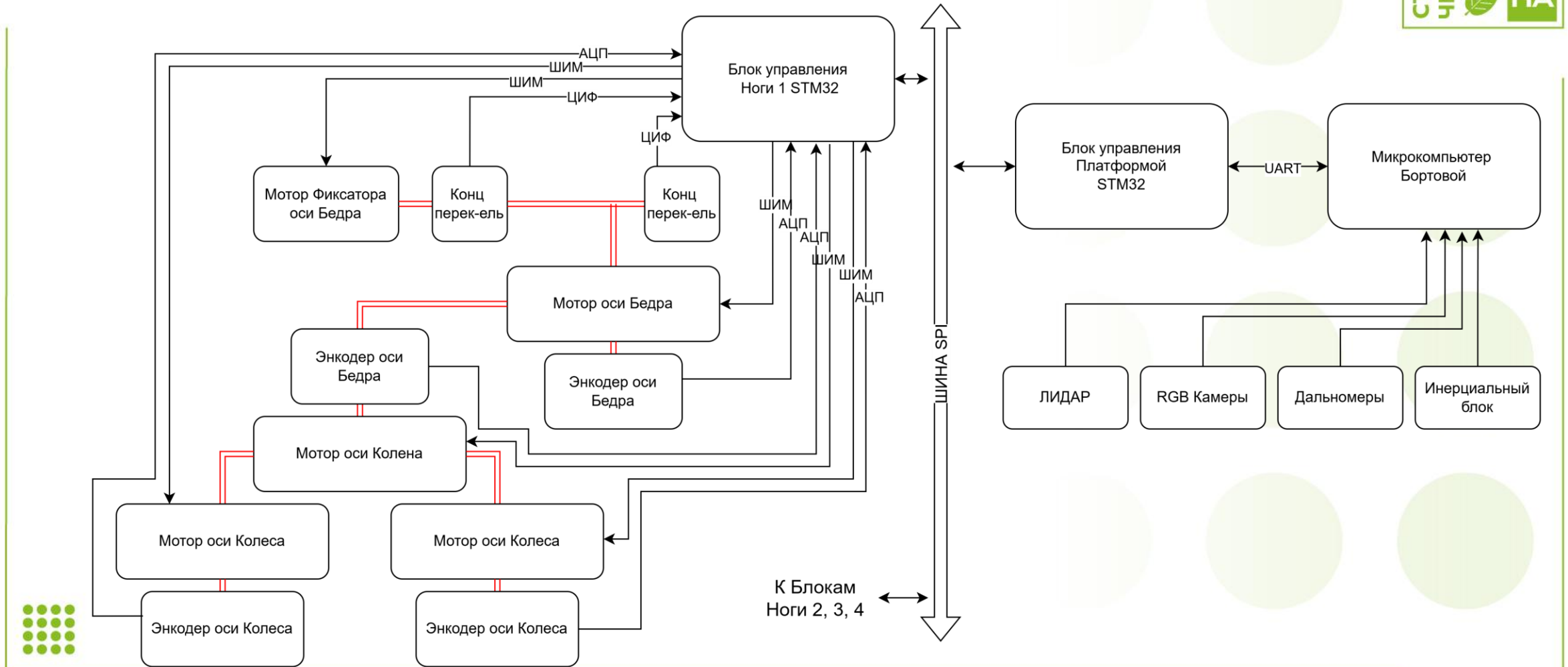


Fig. 16. Platform Control Functional diagram

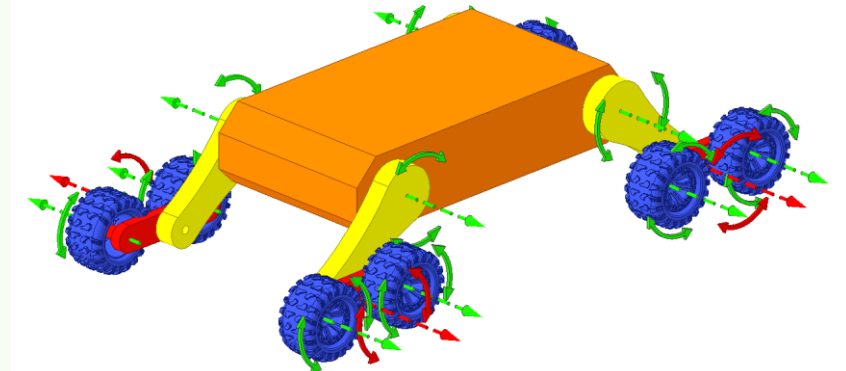


RESULTS



Key Results:

- Platform kinematics model developed.
- Control system architecture designed.
- Cross-country performance validated against technical specifications.





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АПРЕЛЯ

